PassFrame tuition

Perimeter & Area

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A rectangle is 9 cm long and 6 cm wide. What is its area and perimeter?
Method & answer: Area = 9 × 6 = 54 cm². Perimeter = 2 × (9 + 6) = 30 cm.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangle is 7 cm by 5 cm. What is its perimeter?

2. A square has sides of 6 cm. What is its perimeter?

3. Which of these has an area of 44cm²? A) an 11cm×4cm rectangle B) a 7cm×6cm rectangle C) a 6cm×6cm square D) a 9cm×5cm rectangle

4. A square has sides of 8 cm. What is its area?

5. A rectangle is 13 cm long and 5 cm wide. What is its perimeter?

6. A square has a perimeter of 32 cm. What is the length of one side?

7. A rectangle is 9 cm by 6 cm. What is its area?

8. A square has an area of 64 cm². What is the length of one side?

9. Select ALL of these shapes that have a perimeter of exactly 34cm: A) a 12cm×5cm rectangle B) a 10cm×7cm rectangle C) a square of side 8.5cm D) a 9cm×8cm rectangle E) a 13cm×4cm rectangle

10. A rectangle has a length of 14 cm and a width of 5 cm. What is its perimeter?

11. A square has sides of 10 cm. What is its area?

12. A rectangle is 16 cm long and 3 cm wide. What is its area?

13. Which is greater: the perimeter of a 9cm×7cm rectangle, or the perimeter of an 8cm square? A) the rectangle B) the square C) they are equal D) cannot tell

14. A square has sides of 13 cm. What is its perimeter?

15. A rectangle has an area of 54 cm² and a width of 6 cm. What is its length?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A triangle has a base of 12 cm and a height of 5 cm. What is its area?

2. A parallelogram has a base of 11 cm and a height of 5 cm. What is its area?

3. Which triangle has an area of 30cm²? A) base 10cm, height 6cm B) base 12cm, height 5cm C) both A and B D) neither

6. A triangle has an area of 36 cm² and a base of 9 cm. What is its height? A) 4cm B) 6cm C) 8cm D) 9cm

7. A rectangle is 2 m long and 60 cm wide. What is its area in cm²?

8. Select ALL of these that have an area of MORE than 40cm²: A) a triangle base 12cm height 8cm B) a parallelogram base 9cm height 5cm

11. A rectangle's length is three times its width. If the width is 4 cm, what is the perimeter?

12. A triangle has a base of 16 cm and a height of 5 cm. What is its area?

13. A rectangle has a perimeter of 46 cm and a length of 16 cm. What is its area?

C) a triangle base 14cm height 6cm D) a 6cm×7cm rectangle E) a triangle base 8cm height 10cm

4. An L-shape is a 7cm×5cm rectangle with a 3cm×2cm square cut from one corner. What is its area?

5. A rectangular field is 50 m by 20 m. What is its perimeter?

9. Two rectangles, 6cm×4cm and 3cm×2cm, are joined with no overlap. What is the total area?

10. A square lawn has a perimeter of 60 m. What is its area? A) 225m² B) 200m² C) 196m² D) 256m²

14. An L-shaped garden is a 15m×7m rectangle with a 6m×2m rectangle removed from a corner. What is its area?

15. A 24cm×18cm photo is placed in a frame with a 3 cm border all around. What is the total area of the framed photo?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangle has a perimeter that is double the perimeter of a 4cm×3cm rectangle. What is the new rectangle's perimeter?

2. A square's area is 9 times a 4cm² square. What is the side length of the larger square? A) 6cm B) 12cm C) 36cm D) 4cm

3. Two identical rectangles together have a total perimeter of 50cm. What is the perimeter of ONE rectangle?

4. Select ALL of the following that would make a 6cm×4cm rectangle's area double: A) doubling the length only B) doubling both length and width C) doubling the width only D) adding 24cm² to the area E) tripling the length

5. The perimeter of a square is the same as the perimeter of a 10cm×6cm rectangle. What is the area of the square?

6. A rectangular field is 60 m by 35 m. A path 2 m wide is built all the way around the inside edge. What is the perimeter of the inner rectangle (after the path is removed)?

7. Three square tiles, each with side 15 cm, are placed in a row touching each other. What is the perimeter of the resulting rectangle?

8. A rectangular room measures 6 m by 4.5 m. Carpet costs £14 per m². What is the total cost of carpeting the room?

9. A rectangle's length is twice its width, and its perimeter is 42cm. Select ALL of these that are TRUE: A) the width is 7cm B) the length is 14cm C) the area is 98cm² D) the width is 14cm E) the area is 84cm²

10. A large rectangle 16m×10m has a 6m×4m rectangle removed from a corner, forming an L-shape. What is the perimeter of the L-shape?

11. A rectangular sports pitch is 90 m long and 45 m wide. A running track goes around the outside with a uniform width of 4 m. What is the area of the track only (not including the pitch)?

12. A parallelogram-shaped field has a base of 40 m and a height of 25 m. If 1 hectare = 10,000 m², what is the field's area in hectares?

13. A square and a rectangle have equal areas. The square has sides of 15 cm and the rectangle's width is 5 cm. What is the rectangle's perimeter?

14. An L-shaped room is made of two rectangles, 7m×5m and 3m×2m (no overlap). Carpet costs £16 per m², plus a £30 fitting fee. What is the total cost?

15. A rectangular banner's length is tripled and its width stays the same. If the original area was 18 cm², what is the new area?